

Research Paper Structure

Product Optimization & Revenue Contribution Analysis for Afficionado Coffee Roasters

This document provides the complete academic research paper structure for the Afficionado Coffee Roasters Product Optimization & Revenue Contribution Analysis project.

Title Page

Include project title, author name, organization, submission date, and institutional details.

Abstract

Summarize the project objectives, methodology, analytical techniques, major findings, and business impact in approximately 150–250 words.

1. Introduction

Discuss background information about specialty coffee retail, menu optimization challenges, operational complexity, and the importance of product-level analytics.

2. Problem Statement

Explain the business challenges including lack of product profitability visibility, menu inefficiency, revenue concentration risks, and operational dependency issues.

3. Research Objectives

Define primary and secondary objectives including product popularity analysis, revenue contribution analysis, Pareto analysis, and menu optimization goals.

4. Literature Review

Review previous studies related to retail analytics, menu engineering, Pareto principle applications, coffee retail optimization, and business intelligence systems.

5. Dataset Description

Describe the transactional dataset including variables, data structure, store information, product hierarchy, and transactional granularity.

6. Methodology

Explain the analytical workflow including data cleaning, validation, revenue computation, exploratory data analysis, KPI generation, and visualization techniques.

7. Product Popularity Analysis

Analyze product demand patterns, units sold, top-selling products, least-selling products, and category-level popularity distribution.

8. Revenue Contribution Analysis

Measure revenue generated by products and categories, revenue share percentages, premium product contribution, and profitability distribution.

9. Pareto Analysis

Evaluate revenue concentration patterns to identify whether a small number of products generate the majority of total revenue.

10. KPI Construction

Develop key performance indicators including total revenue, average order value, revenue concentration ratio, and product efficiency metrics.

11. Dashboard Development

Describe the Streamlit dashboard development process, interactive features, filters, KPI cards, charts, and downloadable reporting capabilities.

12. Results and Findings

Present analytical findings supported by charts, tables, product rankings, revenue distributions, and operational insights.

13. Discussion

Interpret the business implications of the findings including operational efficiency, menu engineering, product portfolio management, and merchandising strategy.

14. Recommendations

Provide actionable recommendations including menu optimization, inventory improvements, revenue growth strategies, and operational simplification.

15. Conclusion

Summarize the overall project outcomes, business value, and importance of data-driven product analytics in specialty coffee retail operations.

16. Future Scope

Discuss future opportunities such as forecasting models, recommendation systems, customer segmentation, and predictive analytics.

17. References

List academic papers, retail analytics studies, menu engineering literature, business intelligence resources, and industry reports.

Appendices

Include additional charts, dashboard screenshots, code snippets, KPI tables, data dictionaries, and supplementary analysis outputs.

Suggested Visualizations

Visualization Type	Purpose
Bar Charts	Top-selling and top-revenue products
Pie Charts	Category revenue distribution
Scatter Plots	Popularity vs revenue relationship
Pareto Charts	Revenue concentration analysis
Heatmaps	Store-level product performance
Treemaps	Revenue hierarchy visualization
KPI Cards	Executive performance monitoring